Microwave Optics and Bragg Diffraction in Cubic Crystalls

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Characteristic properties of microwaves are studied with the Gaussian radiation distribution of a Gunn diode. Using the periodicity of standing waves and a Michelson interferometer, the wavelength is determined. The transmitted signal is linear polarized, however, due to reflection and diffraction at the experimental setup the intensity does not follow Malus' law in good approximation. Sending microwaves through a double paraffin prism, the exponential decay of an evanescent wave in the optical thinner medium is verified. Finally, the lattice constant of a styrofoam crystal model is measured with the θ-rocking curve method.

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1 Introduction

The spectrum of electromagnetic radiation covers a wide range of wavelengths, extending from long radio waves down to very short gamma rays. To study optical phenomena in the lab, microwave frequencies on the lower hand of this spectrum are preferred.

Since microwaves transform the scale of the experiment, physical quantities of traditional optics become visible and can easily be manipulated. In this experiment a quantitative analysis of reflection, polarization and tunneling as well as microwave models of the Michelson interferometer and Bragg diffraction in cubic crystals are considered.

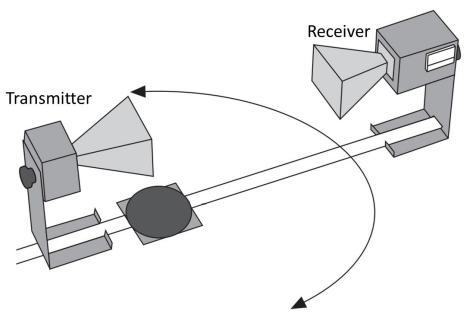

Fig. 1: Signal distribution. The microwave transmitter is centered over the degree plate of the Goniometer, while the receiver scans the intensity dispersion for different angles of rotation [4].

2 Experimental Setup

In every experiment a Gunn diode with a horn of rectangular cross sectional area to direct the output is used as the microwave transmitter, while a Schottky diode with an identical horn collects the emitted signal [3]. Connected to an Agilent 34405A [5] voltmeter, the voltage drop caused by the incident beam is measured. To increase the sensitivity of the detector diode, an operating voltage is applied, why even with the microwave transmitter off, a systematic offset is measured and has to be corrected for each experiment.

Further, the reading of the receiver is not directly related to either the electric field or the intensity of the microwave, rather it's useful for measuring the relative intensity at constant distance, whereby such graphic displays of the sampling data is chosen [3].

3 Dispersion of a Gunn Diode

To examine the radiation pattern of the transmitter, a Goniometer with two arms is used. The long arm is fixed, the other free to rotate around the center of the base. With the receiver directly facing the microwave source and as far back on the rotatable arm as possible, the meter reading at each setting of the angle ϕ is recorded as shown in fig. 1. Due to the specified aperture angle of the attached horn, the radiation penetrates only a narrow spatial area, wherefore the relative intensity in fig. 2 is displayed in a range of $\pm$ 25°.

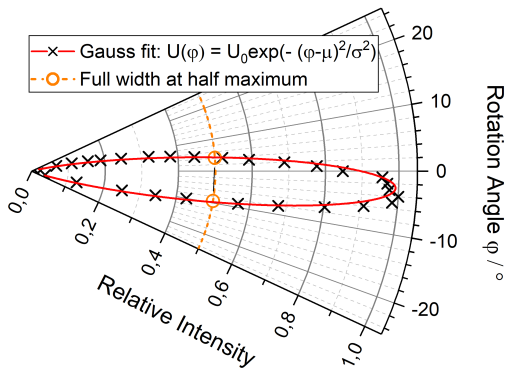

Fig. 2: Gaussian radiation dispersion of the microwave transmitting Gunn diode with a full width at half maximum (FWHM) of $(14,10 \pm 0,26)^\circ$.

As expected, the microwave source has a roughly Gaussian output distribution and emits a strong beam of radiation centered along the axis of the horn. The full width at half maximum (FWHM) is calculated from the standard deviation σ of the Gaussian:

$$\mathrm{FWHM} = 2\sqrt{\ln(2)}\,\sigma, \tag{1}$$

that is with fitted σ and Gaussian error propagation

$$\mathrm{FWHM} = (14.10 \pm 0.26)^\circ$$

The radiation is symmetrical within the limits of statistical fluctuations, but the axis of symmetry is not located on the 0° level. This angular shift of $\mu = (-2.69 \pm 0.10)^\circ$ is put down to the source's alignment on the Goniometer table and is just a systematical error in the experimental setup.

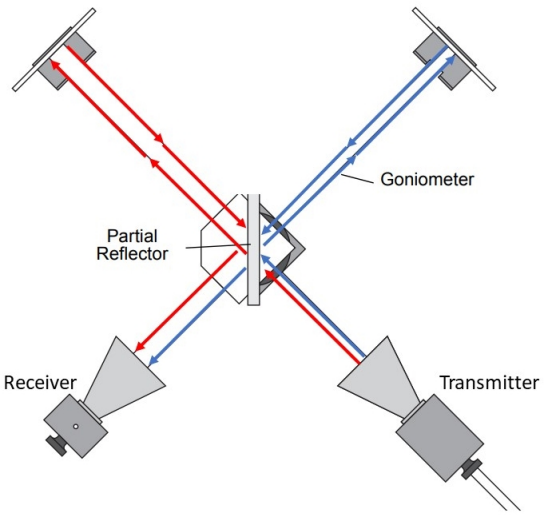

Fig. 3: Michelson Interferometer. A hardboard is used as a partial reflector to split the beam of microwaves. One reflector is fixed, the other adjustable to vary the optical path length and observe interference effects. [4].

4 Wavelength of Microwaves

When two coherent waves meet in space, they superpose, and if they travel in opposite direction, they even form standing waves. In this experiment, microwaves focused by wax lenses are reflected by a metal plate to create a standing wave pattern from which the wavelength λ can be measured directly. That's due to the periodicity of fixed nodes in $\lambda/2$, which density can be detected using a diode. Hence, after n nodes and a traversed distance Δs, the wavelength is determined by

$$\lambda = 2\,\frac{\Delta s}{n} \tag{2}$$

With $n = 35$ nodes after $\Delta s = (55.70 \pm 0.28)$ cm, an estimated reading error of $\sigma = 2$ cm and Gaussian error propagation, the wavelength is calculated to

$$\lambda_\mathrm{s} = (3.276 \pm 0.017)\ \mathrm{cm}.$$

Besides that, the wavelength of the microwave source can be deduced using a Michelson interferometer. According to fig. 3, by splitting and recombining a single emitted wave via a hardboard and metal reflectors, the optical path length and relative phase of two interfering beams is varied. Thus, when one reflector is moved along the Goniometer arm a periodic interference pattern of relative maxima and minima can be observed. Passing through $n = 15$ minima after a traversed distance of $\Delta s = (24.50 \pm 0.14)$ cm, the wavelength is given by

$$\lambda_\mathrm{m} = (3.267 \pm 0.028)\ \mathrm{cm}.$$

In order to verify the compatible measured data the manufacturer's specifications on the microwave transmitter are considered. Given an error-free frequency of $\nu = 93$ GHz, the wavelength is determined to $\lambda_\mathrm{lit} = 3.2$ cm. Because of the temperature sensitive spectrum of Gunn diodes and a lack of significant digits, the experimental data does not agree with the expected value within their uncertainty interval. At least the deviation is system immanent and both measurement methods share the same value, why the actual wavelength of this and the following experimental setups is given by the weighted average

$$\lambda_\mathrm{exp} = (3.272 \pm 0.012)\ \mathrm{cm}.$$

5 Polarization of Microwaves

To examine the polarization properties of microwaves, for starters one can rotate the detector with respect to the transmitter. Being parallel aligned, a maximum in the intensity is measured, while for a relative turn of 90° no radiation can be detected. Thus, it's already determent that the emitted microwaves of the Gunn diode are linearly polarized.

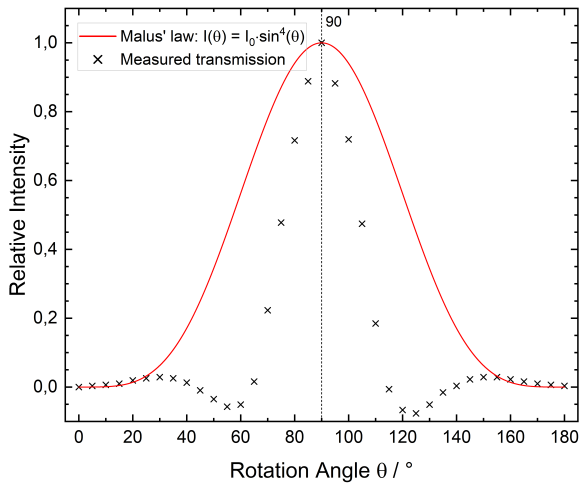

Fig. 4: Transmission intensity through an grid polarization filter at the angle θ to the diode axis of the microwave source and detector. Here, $\theta = 90°$ corresponds to a horizontal polarization orientation.

To support this conjecture, a metallic grid is placed in the ray path. Since its gaps are of the scale of the emitted radiation, the grid works as a polarization filter. An electric field parallel to the metallic rods will be reflected, while in orthogonal orientation it will be transmitted. In a series of measurements with parallel aligned emitter and detector, the filter is rotated to angles θ between $\pm\,90°$. According to Malus' law, the transmitted intensity when passing through the polarization filter is then given by

$$I = I_0 \sin^2(\theta). \tag{3}$$

Since the detector responds only to the component of a microwave signal that is polarized along the diode axis, the detector itself acts as a vertical polarization filter [3]. Hence, the measurement of the relative intensity should show a proportionality of

$$I \sim \sin^4(\theta). \tag{4}$$

However, the experimental intensity distribution in fig. 4 is sharper, which might be due to reflection and diffraction caused by a non-ideal polarization filter. The induced current in the grating bars would not proceed without loss, why the transmitted microwave signal drops faster than theoretically expected.

Transmitter and detector were just chosen to be parallel aligned. When turned orthogonal, the maximum intensity is shifted to a grid orientation of $\theta = 45°$. While without a polarization filter no radiation would be detected, with this setting the component of the electric field vector orthogonal to the polarization axis is transmitted and projected onto the detectors diode. The signal disappears again only for an angle $\theta = 90°$.

6 Focal Length of a Wax Lens

In all optical experiments lenses are essential for beam focusing, why it's necessary to know their focal lengths. Therefore, the intensity of the emitted microwave radiation is measured at a constant object width of $o = (81.45 \pm 0.29)$ cm, while the distance i between detector and lens is varied. That given, the focal length f can be derived from the imaging equation for thin lenses at maximum intensity by

$$f = \frac{o\,i}{o+i}. \tag{5}$$

The effective point of emission and reception for the image and object width is considered according to [3]. As shown in fig. 5, the intensity distribution in the region of the lenses focus has a maximum. However, the data is overlaid with a periodic oscillation pattern of a single standing wave. That might be attributed to the lab's rectangular geometry or to reflections at the detector's horn and the mounting of the wax lens.

To determine the intensity maximum in the image point i, an even polynomial fit of fourth order is applied to the experimental data. This method averages the periodic fluctuations and retains the characteristic intensity dispersion of the lens, why the residuals scatter symmetrically around the optimal fit. Using the polynomial's maximum and it's standard deviation as the image width $i = (70.1 \pm 1.4)$ cm as well as Gaussian error propagation, the focal length is thus calculated to

$$f = (37.68 \pm 0.81)\ \text{cm}.$$

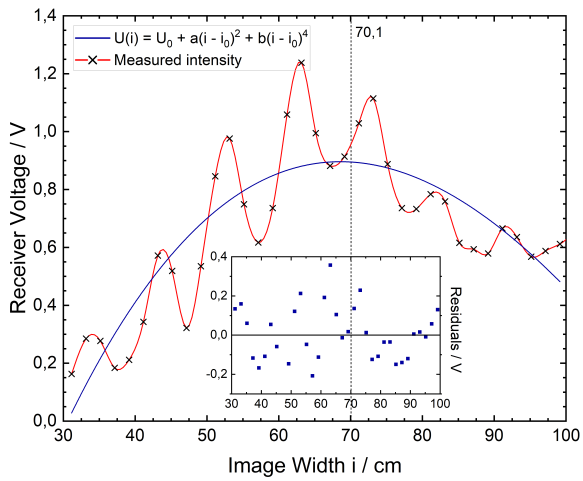

Fig. 5: Microwave intensity distribution in the focus of a wax lens. The data is overlaid with a standing wave pattern, why an even polynomial fit is applied to reduce the noise. At maximum intensity the focal length is determined to $f = (37.68 \pm 0.81)$ cm.

7 Evanescent Waves

In the spirit of geometric optics, radiation is totally reflected at a boundary between isotropic media with refractive indexes n_i above a critical angle

$$\sin(\theta_\mathrm{g}) = \frac{n_2}{n_1}, \qquad n_2 < n_1. \tag{6}$$

In fact, an evanescent wave still propagates into the optical thinner medium with an amplitude that decays exponentially with the distance d to the surface. The intensity at the interface to medium n_2 follows

$$I \sim \mathrm{e}^{-d/\delta}, \tag{7}$$

where δ corresponds to the penetration depth of the evanescent wave. This wave can be captured if a medium with sufficiently high optical density is brought close to the surface where the total reflection takes place. Thus, like quantum tunneling, the radiation passes through an optical barrier and can be detected. In this experiment, frustrated total reflection is observed using a Gunn-diode and two paraffin prisms with adjustable gap distance x, given that:

$$d = x \sin 45° \tag{8}$$

Increasing this quantity, the induced voltage drop of the transmitted and reflected microwave is measured. As can be seen in fig. 6 with a logarithmic scale, the transmission declines with distance, while the reflection increases. In theory, energy conservation with respect to the total intensity should not be violated, but because of reflections in the prism and at the medium transition deviations occur. However, in good approximation the tunneling intensity follows an exponential decay with a penetration depth of

$$\delta = (0.1486 \pm 0.0043)\ \mathrm{cm}.$$

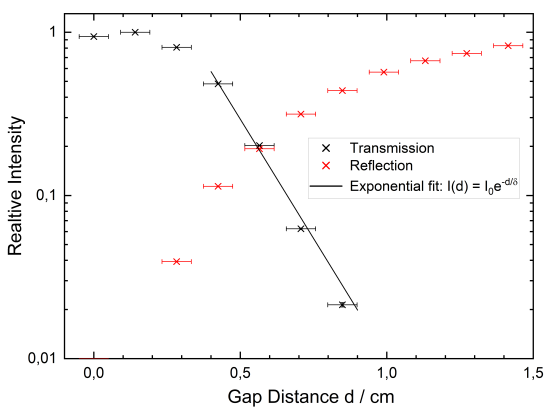

Fig. 6: Transmitted and reflected intensity of an evanescent microwave for two paraffin prisms at distance d. The tunneling intensity decays exponentially with $\delta = (0.1486 \pm 0.0043)$ cm.

8 Bragg Diffraction in Crystals

To study the lattice structure of a cubic crystal with the θ-rocking curve method, x-rays are used since the scale of lattice constants is in the range of Ångstroms. In this experiment, Bragg's law is demonstrated on a macroscopic scale for a styrofoam crystal model.

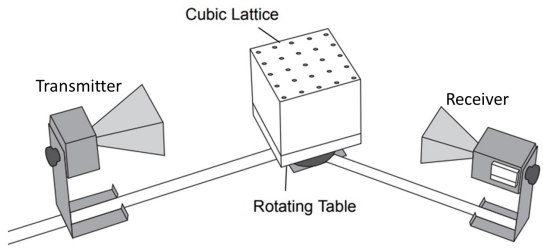

Fig. 7: Bragg diffraction. The lattice constant of a cubic crystal is determined by the θ-rocking curve method. For each plane the crystal is consecutively rotated by θ and the receiver by 2θ [4].

By relating the interplanar spacings d_hkl in the crystal to the scattering angles θ of incident microwaves, the intensity of the diffracted beam is measured for two crystal planes with Miller indices (100) and (110). The lattice constant d_0 can be derived using only Bragg's law for constructive interference at angles θ_m:

$$m\ \lambda = 2\ d_\mathrm{hkl}\ \sin(\theta_m) \tag{9}$$

For cubic crystals the lattice constant then follows to

$$d_0 = \frac{m\ \lambda}{2\ \sin(\theta_m)} \sqrt{\mathrm{h}^2 + \mathrm{k}^2 + \mathrm{l}^2}. \tag{10}$$

Therefore the relative intensity of the diffracted microwave signal is shown in fig. 8/9 as a function of the scattering angle of the incident beam.

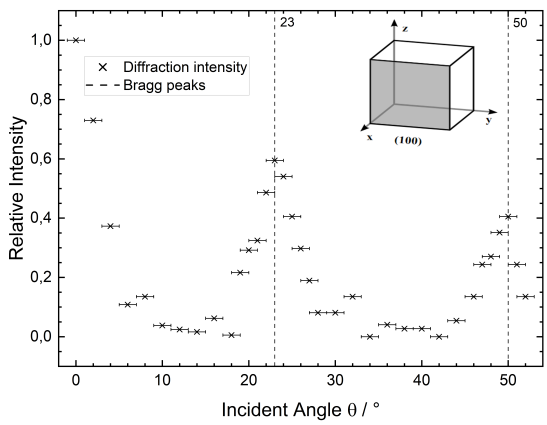

Fig. 8: Intensity distribution of the diffracted microwave signal as a function of the scattering angle θ for the (100) plane [6]. With Bragg's law the lattice constant is determined to $d_0 = (4.261 \pm 0.061)$ cm.

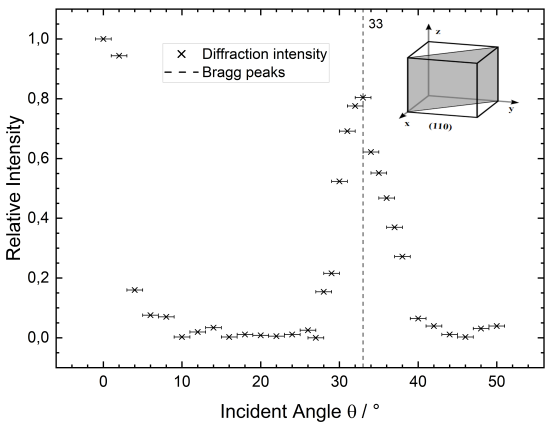

Fig. 9: Intensity distribution of the diffracted microwave signal as a function of the scattering angle θ for the (110) plane [6]. With Bragg's law the lattice constant is determined to $d_0 = (4.25 \pm 0.12)$ cm.

While the (100) plane shows two Bragg peaks for the scanned angular range, the experimental setup allows only the mapping of the first resonance for the (110) plane. However, all values for the lattice constant in tab. 1 are compatible within their margin of error. Given that the reading precision $\sigma_\theta = 1°$ is the same, the confidence increases with the order of maximum, why their weighted average is calculated:

$$d_{\text{exp}} = (4.258 \pm 0.054) \text{ cm}$$

In addition, the lattice constant was directly measured for an open styrofoam block with square configuration as the distance between scattering centers:

$$d_{\text{lit}} = (3.871 \pm 0.050) \text{ cm}$$

Both values do not agree within their uncertainty interval, but at least it can be said that the order of the determined lattice constant is correct. However, it can not be guaranteed for sure that the lattice constant of the crystal model and the open styrofoam block are the same.

plane	order m	angle $\theta/°$	lattice const. d_0/cm
(100)	1	23 ± 1	4.19 ± 0.17
	2	50 ± 1	4.272 ± 0.065
(110)	1	33 ± 1	4.25 ± 0.12

Tab. 1: Bragg diffraction with microwaves in a cubic crystal made of styrofoam. By recording the θ-rocking curve, Bragg peaks at angles θ_{m} are measured and the lattice constant d_0 is determined.

9 Circular Waveguide

Like fiber optics in technology, hollow waveguides are the macroscopic means of transport for microwaves. The closing experiment analyzes their transportation properties by placing microwave source and detector on opposite sites of the metal pipe. Despite its curved shape and rippled surface, the signal is transmitted unobstructed even for sealed ends showing the inner conduction of waveguides. That's due to a vanishing electric field on metallic surfaces, why the signal propagates on the resonator's interior and builds discrete oscillation modes transverse to their transmission direction.

In addition, the remaining polarization of microwaves after the transport in a waveguide is analyzed. With a fixed transmitter the orientation of the receiver was turned relatively, showing a pattern of relative minima and maxima. However, when rotating the transmitter, the polarization direction of the waveguide's output changed accordingly. Hence, the pipe preserves polarization, but not necessarily the axis of polarization.

10 Summary

On the scale of microwaves optical quantities were investigated in this experiment. Measuring the angular intensity distribution of the microwave transmitter, its radiation pattern can be evaluated to be Gaussian shaped with a full width at half maximum of $(14,10 \pm 0,26)°$. According to the manufacturer's specifications the emitted wavelength is $\lambda_{\text{lit}} = 3.2$ cm, however, by counting the nodes of a standing wave $\lambda_{\text{s}} = (3.276 \pm 0.017)$ cm and using a Michelson interferometer $\lambda_{\text{m}} = (3.267 \pm 0.028)$ cm this value could not be confirmed.

Further, using a polarization filter, the radiation was found to be linear polarized, whereby the intensity distribution of a metallic grid was not exactly described by Malus' law. Like the tunnel effect in quantum mechanics, the formation of evanescent waves at medium transitions was observed and the exponential decay in the optical thinner medium with a penetration depth of $\delta = (0.1486 \pm 0.0043)$ cm verified. Finally, the θ-rocking curve for a macroscopic crystal model made of styrofoam was recorded an the lattice constant determined to $d_{\text{exp}} = (4.258 \pm 0.054)$ cm.

References

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